

Ashkan Aghamoali

Mechanical Intelligence Research Group, London South Bank University

iamashkan.github.io/portfolio Dubai, UAE ashkaan.aghamoali@gmail.com linkedin/in/ashkan-aghamoali
github.com/iamashkan

I build **extended-reality applications and digital twins** for engineering: AR interfaces that put simulation results, neural-network predictions and robot state into the room with the engineer, and simulation-first twins running the full stack from an STM32 edge node or a ROS 2 workcell up to a Unity twin and an operations dashboard. Six years with a **bio-inspired structural engineering** group in London – where the XR work began, because a lattice you can morph with your hands is understood faster than one you rotate with a mouse.

EXTENDED REALITY

Six AR applications built at LSBU. They are screen recordings rather than screenshots, and they play on the portfolio: iamashkan.github.io/portfolio

Neural network prediction interface

Unity · ML · C#

An in-scene keypad takes design parameters and returns neural-network predictions of structural deformation as live metrics and charts beside the model.

Parametric control panel

Unity · ARKit

Real-time tuning of a lattice — density, radius, height, thickness — with the geometry rebuilding as the values move.

Bio-inspired structure in AR

Unity · Biomimetics

A biomimetic structure with live controls to morph its geometry and step through design variations.

Step-by-step AR explanation

Unity · 3D

An immersive walkthrough of a bio-inspired structure and its simulation results, anchored in the viewer's own room.

AR co-bot interface

Unity · Robotics

An interface for collaborative-robot visualisation, linking digital simulation to physical automation.

Real-time robotic joint visualisation

Unity · Robotics

A robotic joint rendered live with sensor and data overlays, bridging simulation data and physical motion.

Also: environment-aware AR on Niantic Lightship, segmenting sky, ground, trees and buildings from the camera feed to drive shader behaviour (2024; Niantic has since retired the platform), and a browser-based WebAR face filter in Unity and Needle Engine.

DIGITAL TWIN SYSTEMS

Real-time digital twin generation on iOS

Swift · ARKit · RoomPlan · Core ML

Turns an iPhone into an industrial scanning tool: room geometry captured live with ARKit and RoomPlan, equipment detected and labelled on-device with Core ML and Vision, nameplates read by OCR. Scans export as JSON and glTF, sync to a Node.js and PostgreSQL backend and drive a browser dashboard, while an edge gateway binds each asset to real devices over MQTT with confirmation-gated control.

SCOPE — Measured data is kept strictly separate from AI-inferred data throughout. Under development.

UR5e circular motor recovery cell

ROS 2 Jazzy · MoveIt 2 · Gazebo · Unity

A hardware-ready twin for robotic recovery of end-of-life electric motors from district-cooling plant. A UR5e with a Robotiq 2F-85 gripper sorts eleven separable components across reuse, repair, replace and recycle stations; a decision engine scores each part's health and risk from a Digital Product Passport and sensor stream; MoveIt 2 plans the seventeen-step workflow through ros2_control, with a Streamlit dashboard and Unity twin tracking every decision. Modelled per component rather than per motor, because the real question is the motor size above which a robotic cell beats shredding.

SCOPE — Motion planning, workcell and decision engine are real; perception is currently mocked.

Battery sorting digital twin

Unity · FastAPI · scikit-learn

An operational twin for end-of-life lithium-ion recovery, built against the EU Battery Regulation's mandatory battery passport. A Unity control twin streams each cell's image to a FastAPI backend that classifies form factor, draws a state-of-health profile from NASA PCoE aging data and scores risk; a decision engine then routes every cell to reuse, remanufacture, recycle or quarantine under selectable policy modes.

SCOPE — Simulation-only, on a real RecyBat24 subset and NASA PCoE data. Bounded deliberately: cell chemistry needs XRF or spectroscopy, which no RGB image can substitute for.

Emergency-aware adaptive highway lighting

STM32 · MQTT · Unity 6 · Streamlit

A six-lane smart-motorway lighting system built simulation-first across the full edge-to-cloud stack. The emergency case is the thesis: when an ambulance, police or fire vehicle is detected the system raises a 100% safety corridor ahead of it, in that direction only. Otherwise it holds an eco level and brightens only the luminaires ahead of each vehicle, modelling energy, CO₂, luminaire health and remaining useful life. Telemetry runs over MQTT to an operations dashboard, a five-node STM32 network over RS485 and CAN, and a Unity 6 night-highway twin.

SCOPE — Savings are measured against an always-on baseline — many authorities now switch motorway lighting off overnight, a baseline this does not beat on energy alone.

Also: a defect-inspection twin — PyTorch surface-defect detection with Grad-CAM localisation and ML grading into reuse, repair or recycle, inside a ROS 2 and Gazebo inspection cell.

EXPERIENCE

AI Researcher / XR Developer

2020 — present

Mechanical Intelligence Research Group, London South Bank University — London, UK

- Develop bio-inspired structures within a multidisciplinary group spanning biology, mechanics and computation.
- Built a Python automation system integrated with CAE software for parametric modelling, and analysed over 1,000 Abaqus simulation instances through it.
- Trained machine-learning models of structural behaviour on that corpus, with optimisation and sensitivity analysis feeding back into design tuning.
- Developed the group's XR software for iOS and Android, presenting products, simulations and research findings in AR.

Research Assistant — computer vision

2016 — 2020

University of Guilan — Rasht, Iran

- Built a detection system for camouflaged moving targets in video; a hybrid HSV and Local Binary Patterns approach improved accuracy by 40%.
- Wrote custom CUDA kernels for an 8× speed-up, enabling real-time 4K analysis at 30 fps, and an OpenGL visualisation module with dynamic thresholding.

RESEARCH

S. H. Eraghi, A. Toofani, **A. Aghamoali**, A. Zoroufi, M. Shafaghi, S. Jafari, S. Goel, H. Rajabi. "The rise of metaverse manufacturing: towards a universal, sustainable and bioinspired industrial ecosystem." *Journal of the Royal Society Interface*, vol. 23, 2026. doi:10.1098/rsif.2025.0480

A. Toofani, S. H. Eraghi, **A. Aghamoali**, A. Zoroufi, A. Khaheshi, A. Basti, S. Gorb, H. Rajabi. "One to many: an intelligent biomimetic design framework for functional diversification." In review, 2025.

A. Aghamoali, M. Nahvi. "Improvement of camouflage moving target detection in video sequences." *Journal of Mathematical Modeling*, 2020.

Gold Medal — 3rd International Invention & Innovation Competition, International Federation of Inventors' Associations (IFIA), August 2023.

TECHNICAL

EXTENDED REALITY	Unity 3D, ARKit, RoomPlan, RealityKit, WebXR, Needle Engine, Shader Graph / HLSL, Niantic Lightship, C#, Swift
DIGITAL TWINS	ROS 2, MoveIt 2, Gazebo, ros2_control, MQTT, STM32 and embedded C++, RS485 / CAN, FastAPI, Streamlit, PostgreSQL
MACHINE LEARNING	PyTorch, scikit-learn, computer vision, object detection, Grad-CAM, surrogate modelling on simulation data, CUDA
ENGINEERING & PLATFORM	Abaqus / CAE automation, parametric and optimisation workflows, Python, JavaScript / TypeScript, Node.js, Docker, Cloudflare Workers

CERTIFICATION

Machine Learning Specialization Stanford · Deep Learning Specialization DeepLearning.AI · Deep Neural Networks with PyTorch IBM · AR/VR/MR/XR with WebXR, Unity & Unreal Michigan · Digital Technologies & the Future of Manufacturing Michigan · Mobile AR Development and Unity Essentials Unity, verified on Credly

EDUCATION

MSc, Digital Electronic Systems <i>University of Guilan, Iran · 2016 – 2019</i> <i>English – professional · Persian – native</i>	BSc, Electrical & Electronics Engineering <i>University of Guilan, Iran · 2012 – 2016</i>
---	---